Undersigned: talking points for Alyssa

Fri 30 Oct · MS3 due Thu 12 Nov · each item is what we wanted, what we tried, what happened

1 Linking comments to the right section

WE WANTED
MS2 could only place comments that cite a section number, and only 3.7% do. We wanted a model that reads any passage and says which section of the rule it's arguing about.

WE TRIED
Fine-tuning a small embedding model (bge-small). Labels come free: take comments that do cite, hide the citation, make the model guess it. Trained as a Vertex job, logged in W&B.

RESULT
65.5% right on the first guess over 490 test passages. Adding two extra signals (the agency's own discussion of each section, and other sections the letter cites) brought it to 66.7%. On 40 hand-labelled passages with no citation: 90%.

2 How it compares to an LLM

WE WANTED
To know whether a free small model is good enough, or whether we just need to pay for Claude. Same 232 passages for everyone.

RESULT

Method	Right first	$ per 1,000
Plain search (MS2)	53.4%	0
Our model	**69.8%**	**0**
Haiku	69.0%	1.29
Sonnet	75.4%	19.58

We tie Haiku and are about 6 points behind Sonnet.

3 Getting close to Sonnet without paying for Sonnet

WE WANTED
Sonnet-level accuracy at a cost we could afford on the full data. Running Sonnet on all 721k passages would cost $14,121.

WE TRIED
A cascade. Our model answers when it's confident and hands the unsure ones to Sonnet. This works because its confidence means something: on its surest fifth it beats Sonnet (96% vs 87%), on its least sure fifth it loses badly (33% vs 67%).

RESULT
72.0% at $2.43 per 1,000, which is 95% of Sonnet's accuracy for an eighth of the price. On all 721k passages: about $1,750 instead of $14,121. A cheaper setting gets 70.7% for about $960.

4 Rules the model has never seen

WE WANTED
The model to work on a new rule out of the box, since every campaign is about a different rule.

WE TRIED
Holding out one rule (ED) entirely, then adding the agency's preamble text to training so the model learns how agencies describe their own sections.

RESULT
On the unseen rule: 50% before, 67% with the preamble. Plain search gets 57% there.

5 Putting it in production

WE WANTED
The MS3 stack: Vertex training, W&B, a Vertex pipeline, and the model served on Cloud Run.

WE TRIED
A pipeline that trains, evaluates, and only replaces the live model if the new one beats both plain search and the current model. Then deployed to Cloud Run with logging and a `/stats` page.

RESULT
The new model passed (+12.9 points over search) and is live. Live check on 80 LCRI passages: 57.5% right first, 88.8% in the top 5, 172 ms typical response. 44 API tests run in CI.

6 GPU training

WE WANTED
To train on GPUs, which would also cover the optional multi-GPU part of MS3.

WE TRIED
Requesting 2 T4 GPUs in us-central1 on 29 Oct.

RESULT
Google blocked it because the project is on a free-trial billing account. It needs the billing owner to upgrade to paid. So far we've trained on CPU, which has been fine.

7 Where it's still weak

- Two rules are bad: CPP 46.7% and LCRI 52.5%. The others are 60 to 76%.
- The labels limit us. Two of us labelled the same 30 passages and agreed on only 21 (70%), so the ceiling is around 83%.
- The live endpoint still uses only the base model. The two extra signals from item 1 aren't served yet.
- We've only held out one rule. The other five haven't been tested as unseen.

8 Questions for Alyssa

- Is CPU training on Vertex OK, or does the course have a billing account we should use for GPUs?
- The Cloud Run service is private. Does it need to be public for grading, or is staff access enough?
- Is it fine to lead with the cascade as our main result, or should we lead with the plain model vs. Sonnet?
- Do you want unseen-rule tests on the other five rules too, or is one enough?